Stockout-Aware Demand Forecasting and Newsvendor Ordering on FreshRetailNet-50K

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***Abstract*—Recorded sales understate demand whenever a product sells out, so forecasts and orders fitted to sales inherit the stocking policy. We study demand forecasting and newsvendor ordering on FreshRetailNet-50K, a fresh-retail panel with hourly stockout flags. On days without a flagged stockout, recorded sales give an exact, though selected, proxy for demand. We censor these days with three simulated stockout mechanisms, so that models trained on differently censored histories can be scored against the same uncensored outcomes. In a 5,000-series run whose protocol was locked before the final week was scored, we compare nine ordering policies under three cost ratios, and no policy wins under all of them. When a shortage costs three times as much as a leftover unit, LightGBM demand quantiles with pooled conformal calibration are the cheapest full-sample policy in all nine train/test cells; when leftovers cost more, a raw-sales LightGBM with residual quantiles is cheapest in all nine. Training on a different mechanism raises the calibrated policy's mean cost by at most 0.0472 normalized units (95% cluster-bootstrap interval 0.0391–0.0554), the smallest worst case among the policies that read the evaluation history. For real censored days, where demand is unobserved, we report only a lower bound on misses of the 90% quantile forecast.**

***Index Terms*—censored demand, stockouts, demand forecasting, quantile regression, conformal calibration, newsvendor, fresh retail**

I. Introduction

Retailers record sales, not demand. When a product sells out, the customers who could not buy it leave no trace in the data [1]. A model fitted to such records learns the stocking policy along with customer behavior, and in fresh retail its errors go straight into the next order, which has to balance lost sales against perishable surplus [2], [3]. A lower forecast error is therefore not enough to show that one ordering policy is better than another. A fair comparison needs the same outcome, the same cost ratio, and the same scored rows.

FreshRetailNet-50K [4], [5] makes such a comparison possible because it stores an hourly stock-status flag next to hourly sales. We call a series-day with no flagged stockout between 06:00 and 21:59 a *donor day*; if the flags are correct, its recorded sales over that window equal demand. Donor days are not a random sample, and they tell us nothing about days that were actually censored. They do, however, allow a controlled experiment, much like the recovery evaluation in the dataset report [4]. We hide part of the sales on donor days with three simulated stockout mechanisms, rebuild features and forecasts from each censored history, and score every forecast and order against the uncensored proxy. A model trained under one mechanism can then be tested under another on exactly the same rows.

The results come from three saved runs of one notebook pipeline (Fig. 1). Two development runs prepare the data, simulate censoring, and tune the models: run S on a stratified sample of 5,000 series and run P on all 50,000 series. Run E repeats these steps on its own 5,000-series sample and then adds the neural and Kaplan–Meier baselines, calibration, ordering, a protocol lock, and a single evaluation of the final week. Only run E is used in the final evaluation.

The paper contributes an evaluation design and the evidence it produces rather than a new forecasting model. Earlier work on this dataset reports demand recovery and forecast accuracy [4] or profit measured on censored sales [6]. First, we score ordering cost against an uncensored proxy, so that nine policies are compared on identical rows. Second, we vary how the training history was censored and measure what it costs to move a model between mechanisms, with paired cluster-bootstrap intervals. Third, we show that the ranking of calibrated demand quantiles and a raw-sales model reverses with the cost ratio in every train/test cell, and we bound the miss rate of the 90% quantile forecast (q90) on real censored days.

II. Related Work

Classical methods for censored demand fit a parametric distribution to the observed sales [2] or learn order levels nonparametrically with the Kaplan–Meier estimator [3], [7]; our Kaplan–Meier baseline follows the second route. Besbes and Muharremoglu [8] show what censoring hides about the optimal order, Sachs and Minner [9] adapt the data-driven newsvendor to censored samples, and Kumar and Mouchtaki [10] show that treating sales as demand can degrade badly as censored data

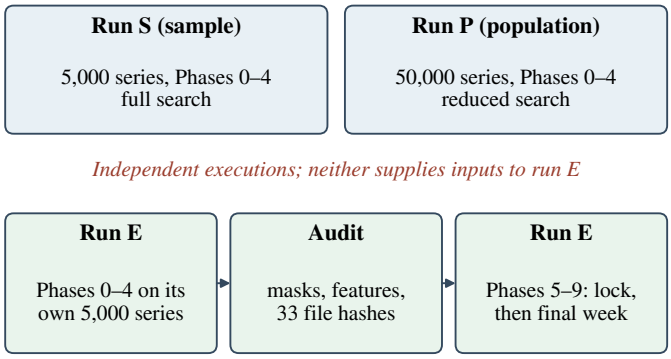


Fig. 1. The three saved runs. Runs S and P are independent development runs; only run E, which repeats Phases 0–4 itself, feeds the locked final evaluation.

accumulate. Feature-based and neural newsvendor methods learn orders directly from covariates [11]–[13]. With linear costs the optimal order is a quantile, which links ordering to quantile regression [14].

LightGBM [15] was used by practically all of the top 50 entries in the M5 accuracy competition [16], and simple models often match transformer forecasters [17]. We also use TimesNet [18] to recover masked hours and the Temporal Fusion Transformer (TFT) [19] for next-day forecasts. Our calibration builds on conformalized quantile regression [20], adaptive conformal inference [21], and monotone rearrangement [22], and Cao [23] applies conformal calibration to the feature-based newsvendor. These guarantees assume exchangeability or bound the departures from it [24], and real censored days satisfy neither. On FreshRetailNet-50K itself, the dataset report [4] evaluates recovery on synthetically masked non-stockout periods with an official baseline [25] that we could reproduce only in part, and Azar [6] selects newsvendor policies but scores profit on censored sales, whose bias relative to demand cannot be identified. Other stockout-aware studies leave out-of-stock days out of training and evaluation [26] or feed a global forecaster into a cost-aware ordering rule [27]. None of them scores orders against an uncensored outcome while varying the censoring mechanism.

III. Data and Experimental Protocol

A. Data, outcome, and split

The training file has 4,500,000 series-day rows for 50,000 (city, store, product) series, and the evaluation file has 350,000. The training file holds 865 product identifiers, the number given on the current dataset card [5], while the dataset report states 863 stock-keeping units [4]. We use release 1.0 (CC BY 4.0). Its upstream revision and our download date were not recorded, so the files are identified by the SHA-256 hashes listed at the end of the paper. The outcome is the recorded sales between 06:00 and 21:59, the window that a next-day order has to cover. A series-day is a donor day when its stock-status flags $s_{j,t,h}$ are all zero over that window, and only then do its window sales serve as the proxy:

$$I^{\text{donor}}_{j,t} = \mathbf{1}\Big\{\sum_{h=6}^{21} s_{j,t,h} = 0\Big\}, \quad D^{\text{proxy}}_{j,t} = \sum_{h=6}^{21} y^{\text{sale}}_{j,t,h}. \tag{1}$$

Donor days are a selected sample. In run S, fit-period donor days had a two-week pre-day mean of recorded sales 0.91 times that of the other days, and they fell on promotion days less often (0.35 against 0.39). Real final-week rows with a flagged stockout have no demand label and are never scored for error or cost.

The pipeline runs in ten phases: data checks and sampling (Phase 0), audit and past-only features (1), simulated censoring (2), conventional baselines (3), LightGBM demand quantiles (4), TimesNet and TFT (5), Kaplan–Meier (6), conformal calibration (7), ordering and the protocol lock (8), and the final evaluation (9). All runs use the same chronological split. Days 1–60 (2024-03-28 to 2024-05-26) are used for fitting, days 61–75 (to 2024-06-10) for choosing hyperparameters, and days 76–90 (to 2024-06-25) only for calibration; days 91–97 (2024-06-26 to 2024-07-02) form the final week. The sampling tiers (first-level category, sales tier, and stockout-rate tier) and the initial simulator rules use fit-period data only. Features use only earlier information: target-day flags, sales, weather, and promotions are excluded, lags are shifted by at least one day, and randomizing every target-day field leaves the features unchanged. Until Phase 4 the evaluation file was read only for its schema, identifiers, and dates. Run E wrote its protocol lock in Phase 8, before any final-week outcome was scored, and the final week was used once.

B. Simulated censoring mechanisms

Each mechanism replaces the visible history of eligible donor days (those with enough past data) with a simulated censored one; real censored days keep their recorded values. Under mechanism m, the opening stock of series j on day t is

$$Q^{(m)}_{j,t} = \max\Big\{\mu^{\text{past}}_{j,t}\, k^{(m)}_{j,t} \exp(\sigma_m z_{j,t}),\ q_{\text{floor}}\Big\}, \tag{2}$$

where $\mu^{\text{past}}_{j,t}$ is the mean recorded window sales over the previous two weeks, $k^{(m)}_{j,t}$ a stock multiplier fitted to fit-period stockout patterns, $z_{j,t}$ a seeded standard normal draw shared by all mechanisms, and q_{floor} a small floor. Visible hourly sales are the smaller of proxy sales and remaining stock. Mechanism A (ordinary depletion) may add one refill of half the opening stock one to three hours after the first simulated stockout, with a probability estimated from real within-day recoveries. Mechanism B (promotion-sensitive) multiplies A's stock factor by 0.6 on recorded promotion days; the promotion flag drives the simulator but is not used as a forecasting feature. Mechanism C (sustained scarcity) halves A's multiplier, uses a fixed noise scale, and never refills (Fig. 2). A transfer cell pairs a training mechanism with an evaluation history. Within a cell, every forecast is scored against the same proxy, so cost differences between policies are paired.

TABLE I
MEAN NEWSVENDOR COST OF EVERY LOCKED POLICY FOR A-TRAINED MODELS ON THE A HISTORY (LOWEST COST PER PANEL AND SETTING IN BOLD)

Policy	Population	n	Mean cost $\beta = 0.25$	$\beta = 0.50$	$\beta = 0.75$
(a) Full-sample exact-donor candidates					
Seasonal naive + residual quantile	Full sample	20,641	0.6759	0.4430	0.7853
Raw-sales LightGBM + residual quantile	Full sample	20,641	**0.5448**	**0.3868**	0.7366
Raw-sales quantile LightGBM	Full sample	20,641	0.5749	0.3999	0.7503
Demand-quantile LightGBM (uncalibrated)	Full sample	20,641	0.5816	0.3946	0.7152
Demand-quantile LightGBM + pooled conformal	Full sample	20,641	0.5766	0.3891	**0.6831**
Kaplan–Meier order percentile	Full sample	20,641	0.8404	0.6520	1.3878
Demand-quantile + adaptive calibration (sensitivity)	Full sample	20,641	0.5815	0.3932	0.6963
(b) Declared deep subset of 600 series (separate population)					
TFT (direct) + residual quantile	Deep subset	2,510	0.6892	0.4949	0.9483
TimesNet→TFT + residual quantile	Deep subset	2,510	**0.6586**	**0.4687**	**0.8759**

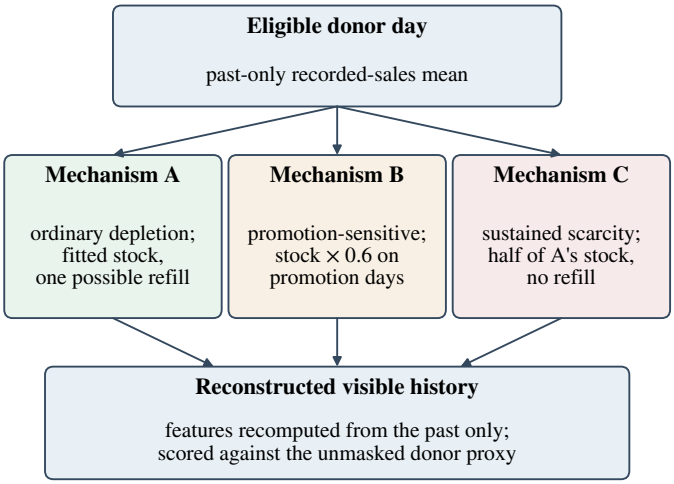


Fig. 2. The three censoring mechanisms act on eligible donor days. Each produces its own visible history, from which past-only features and forecasts are rebuilt; real censored days keep their recorded values.

C. Models, calibration, ordering, and metrics

The conventional baselines are a seasonal-naive rule and LightGBM [15] point and quantile models trained on visible sales. The demand-quantile models are LightGBM quantile regressions trained on the donor proxy at $\tau \in \{0.25, 0.50, 0.75, 0.90\}$, one set per mechanism. On a declared deep subset of series, TimesNet recovers artificially hidden hours, and TFT forecasts the next day either from the visible history (direct) or from the TimesNet-recovered one. The Kaplan–Meier policy estimates an order percentile from censored sales within pools that fall back from category by sales tier to category, management group, and finally a global pool. Pooled conformal calibration shifts each quantile by a residual quantile computed on calibration-window donors in the same pool hierarchy (minimum pool size 200) and then sorts each row's corrected quantiles. An adaptive variant [21] is included as a sensitivity analysis. Point forecasters become quantile forecasts by adding the pooled quantile of their calibration residuals. Table I lists the nine locked ordering policies built from these models.

With a cost c_u per unit short and c_o per unit left over, the newsvendor order is the $\beta = c_u/(c_u + c_o)$ quantile, and the cost of a row is

$$C(Q, D^{\text{proxy}}) = c_o \max(Q - D^{\text{proxy}}, 0) + c_u \max(D^{\text{proxy}} - Q, 0), \tag{3}$$

evaluated against the unmasked proxy under a one-day shelf life. We lock three settings, $(c_u, c_o, \beta) = (1, 3, 0.25)$, $(1, 1, 0.50)$, and $(3, 1, 0.75)$, and call them excess stock more costly, equal costs, and shortage more costly. Costs are in normalized sales units. On donor rows we also report the weighted absolute percentage error (WAPE) of the q50 forecast, the empirical coverage of each quantile, and the pinball loss [14]. These measures answer different questions, and we do not rank methods across them. For a real final-week row with recorded window sales S_{rec} and q90 forecast U_{90}, demand is at least S_{rec}, so every row with $S_{\text{rec}} > U_{90}$ is a certain miss. The definite-miss rate

$$L_{\text{miss}} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{S_{\text{rec},i} > U_{90,i}\} \tag{4}$$

is therefore a lower bound on the real q90 miss rate. It is not a coverage estimate, nor an estimate of censored demand.

IV. DEVELOPMENT RUNS AND MODEL TRAINING

Run S searched hyperparameters fully on 5,000 series, while run P used a reduced search on all 50,000. The two runs have almost the same donor share (0.5580 and 0.5573), and their simulator summaries agree to within 0.001; in run S, for example, mechanisms A, B, and C mask 0.460, 0.566, and 0.832 of the eligible donor days. Run P has a lower q50 tuning WAPE (0.3124–0.3161 against 0.3334–0.3361 over the three mechanisms) and took longer to run, but because the runs differ in search mode as well as in size, neither difference can be put down to scale.

Run E did not reuse run S. It ran Phases 0–4 again with the full search and reproduced run S's selected series, split, simulator masks, donor counts, and q50 tuning WAPEs (0.333440, 0.336140, and 0.335759) exactly, as expected from

TABLE II
PRE-SPECIFIED PAIRED CONTRASTS FOR THE POOLED-CONFORMAL POLICY OVER THE THREE MATCHED HISTORIES (NEGATIVE VALUES FAVOR POOLED CONFORMAL)

Pooled conformal minus	Cost setting	Population	n	Difference	95% percentile interval
Uncalibrated demand quantile	Excess stock more costly	Matched full sample	61,923	−0.0052	[−0.0068, −0.0034]
Uncalibrated demand quantile	Equal costs	Matched full sample	61,923	−0.0077	[−0.0096, −0.0062]
Uncalibrated demand quantile	Shortage more costly	Matched full sample	61,923	−0.0420	[−0.0483, −0.0365]
Raw-sales LightGBM + residual	Excess stock more costly	Matched full sample	61,923	+0.0208	[0.0016, 0.0432]
Raw-sales LightGBM + residual	Equal costs	Matched full sample	61,923	−0.0091	[−0.0204, 0.0052]
Raw-sales LightGBM + residual	Shortage more costly	Matched full sample	61,923	−0.0790	[−0.0946, −0.0615]
Seasonal naive + residual	Excess stock more costly	Matched full sample	61,923	−0.1213	[−0.1465, −0.0876]
Seasonal naive + residual	Equal costs	Matched full sample	61,923	−0.0694	[−0.0872, −0.0439]
Seasonal naive + residual	Shortage more costly	Matched full sample	61,923	−0.1376	[−0.1723, −0.0902]
Kaplan–Meier percentile	Excess stock more costly	KM-supported rows	61,523	−0.4043	[−0.4276, −0.3824]
Kaplan–Meier percentile	Equal costs	KM-supported rows	58,183	−0.3223	[−0.3429, −0.3046]
Kaplan–Meier percentile	Shortage more costly	KM-supported rows	47,294	−0.7475	[−0.8175, −0.6812]

a deterministic pipeline with the same data and seed. Before its later phases, an audit rebuilt the simulator masks and feature matrices from the raw panel, found exact agreement, checked the hashes of 33 saved files, and confirmed that the calibration and final-week days were still unused. Run E's later phases used 200 bootstrap replicates, and all results from here on come from its 5,000 series. The neural models were trained on CPU, with two logical CPUs and four epochs per model, on a declared subset of 600 series. With this budget they are reference points, and we compare them only with each other. TimesNet scored 290,397 artificially hidden donor hours with a WAPE of 0.917 to 0.934 across the mechanisms. For TFT, the next-day WAPE on 5,407 matched tuning donor days falls from 0.451–0.453 with the visible history to 0.413–0.415 with the TimesNet-recovered history. The two tasks have different targets and units, and neither WAPE is comparable with the Phase 4 WAPE. We could reproduce the official baseline [25] only in part: the as-documented TimesNet run finished only after a recorded patch for a blocked network download, and the upstream TFT prediction step was not run.

The Kaplan–Meier implementation passed all 8 of its unit tests and built 207 pools. When a pool cannot support the requested percentile, the order is left unestimated rather than borrowed from another policy. Calibration used 513,276 residuals from the calibration window, and 0.061 of the calibration rows were corrected from a pool coarser than category by sales tier. Sorting removed every remaining quantile crossing, which makes the quantiles ordered but not necessarily calibrated. The order and cost arithmetic passed 10 hand-computed checks. The Phase 8 lock then fixed the policy names, cost settings, model and calibration files, planned comparisons, and bootstrap settings, all before Phase 9 read any final-week outcome.

V. LOCKED FINAL-WEEK EVALUATION

The final evaluation scores donor rows only. Each of the nine transfer cells (one per pair of training mechanism and evaluation history) contains the same 20,641 final donor series-dates, and the six full-sample policies other than Kaplan–Meier are scored on all of them. Kaplan–Meier is scored only where its order is estimable, on 6,012 to 20,641 rows per cell, and the two neural policies on the 2,510 final donor rows of the deep subset. We impute no predictions for the remaining 4,400 series.

A. Policy comparison

Table I lists the mean costs of the A-trained models on the A history. Which full-sample policy is cheapest depends on the cost ratio. The raw-sales LightGBM with residual quantiles wins when excess stock is more costly and, narrowly, at equal costs (0.3868 against 0.3891 for pooled conformal), whereas the pooled-conformal policy wins when shortage is more costly. The B history gives the same ranking. On the C history, pooled conformal is already the cheaper of the two at equal costs (0.4094 against 0.4411). Kaplan–Meier is the most expensive full-sample policy in every setting. In the deep subset, TFT on the recovered history costs less than direct TFT in every matched history and setting.

Table II reports the paired contrasts fixed in the lock, pooled over the three matched histories. Calibration lowers cost compared with the uncalibrated demand quantiles in all three settings, and pooled conformal is cheaper than the seasonal-naive and Kaplan–Meier policies throughout. Against the raw-sales LightGBM with residual quantiles, the sign depends on the cost ratio: pooled conformal costs more when excess stock is more costly, the equal-cost interval contains zero, and it costs less when shortage is more costly. Raising β from 0.25 to 0.75 increases the pooled-conformal fill rate on the A history from 0.614 to 0.891 and the leftover share from 0.056 to 0.226, and the C history is the most expensive in every setting. Calibration moves the q90 coverage of the 20,641 matched rows from 0.798–0.828 to 0.879–0.887 and lowers the q90 pinball loss from 0.119–0.131 to 0.109–0.114. Coverage still falls short of the nominal 0.90, and it says nothing about censored days.

B. Mechanism transfer and robustness

Table III fixes the pooled-conformal policy at $\beta = 0.75$ and varies the history used for training and calibration. The transfer gap $\Delta_{m\to h} = \overline{C}_{m\to h}(\mathcal{I}) - \overline{C}_{m\to m}(\mathcal{I})$ compares the mean cost on an unseen evaluation history h with the matched cost on the

TABLE III
TRANSFER GAPS OF THE POOLED-CONFORMAL POLICY WHEN SHORTAGE IS MORE COSTLY

Train	Test history	Population	n	Matched	Unseen	Gap	95% percentile interval
A	B	Full sample	20,641	0.6831	0.6932	+0.0101	[0.0076, 0.0128]
A	C	Full sample	20,641	0.6831	0.7303	+0.0472	[0.0391, 0.0554]
B	A	Full sample	20,641	0.6940	0.6908	−0.0032	[−0.0059, −0.0006]
B	C	Full sample	20,641	0.6940	0.7306	+0.0365	[0.0275, 0.0447]
C	A	Full sample	20,641	0.7248	0.7120	−0.0128	[−0.0219, −0.0050]
C	B	Full sample	20,641	0.7248	0.7086	−0.0162	[−0.0242, −0.0083]

same rows $\mathcal{I}$. The largest gap occurs for models trained on A and evaluated on C: 0.7303 against 0.6831, a gap of +0.0472. The C-trained models cost less on the A and B histories than on their own, which is the hardest, so a negative gap points to an easier history rather than to better transfer. All intervals are 95% percentile intervals from 200 cluster-bootstrap replicates [28] over the 866 city–store clusters, with the same draws on both sides of a contrast. Dates are not resampled separately and there is no multiplicity adjustment, so we treat the intervals as descriptive.

The cost-ratio reversal holds in all nine train/test cells. When excess stock is more costly, the raw-sales LightGBM with residual quantiles is the cheapest full-sample policy in every cell, with pooled conformal 0.0007 to 0.0323 more expensive. When shortage is more costly, pooled conformal is the cheapest in every cell, 0.0441 to 0.1298 below the raw-sales policy. At equal costs, four cells go to each of the two and one to the uncalibrated demand quantiles. Across the three settings, the largest transfer gap of pooled conformal (+0.0472) is the smallest worst case among the six policies whose forecasts read the evaluation history; the raw-sales quantile LightGBM reaches +0.2258, and Kaplan–Meier orders do not read that history at all. Over the seven final dates (2,507 to 3,385 donor rows each), the C history is the most expensive on every date, and the daily q90 coverage of pooled conformal ranges from 0.839 to 0.907. Under-coverage is concentrated in series with at most ten complete fit-period days (0.752 to 0.769 over 532 rows) and on promotion days (0.854 to 0.865). These breakdowns are descriptive and were not pre-specified.

C. Real recorded histories

We applied the q90 forecasts of the A-, B-, and C-trained pooled-conformal models to the same 35,000 real final-week rows, including rows with flagged stockouts. The definite-miss lower bounds (4) are 0.0895 $[0.0852, 0.0942]$, 0.0854 $[0.0808, 0.0896]$, and 0.0724 $[0.0684, 0.0766]$. A smaller bound can simply come from a higher forecast, so on its own it does not mean better calibration. The bound is highest for rows with 1–4 recorded stockout hours (0.1465 to 0.1797) and lowest for rows with 9–16 hours (0.0298 to 0.0334): when a product sells out early, recorded sales rarely exceed the forecast even when demand does. Only the complete-day stratum has an exact proxy.

VI. DISCUSSION

The reversal follows the direction of the censoring bias. Sales understate demand, which costs little when the order targets a low quantile and a lot when it targets a high one. A model trained on recorded sales therefore does well when excess stock is the expensive error and falls behind when shortage is. This refines the warning that treating sales as demand degrades as censored data accumulate [10]: in our data the penalty appears when shortage is costly. The ablations recorded in run E point the same way. Averaged over the matched histories at equal costs, the pooled-conformal demand quantiles cost 0.3984 against 0.4075 for the raw-sales point model. The adaptive variant cost 0.4039, so its drift correction did not pay off in this week. A hypothetical two-day carry-over of unsold stock lowers the cost to 0.2579, which shows how strongly the cost level depends on the one-day shelf life. Giving the model the target-day weather and promotion values, which are not known when the order is placed, changes the q50 WAPE only from 33.48% to 33.40%.

Limitations. Donor-day sales equal demand only on complete days, and whether a day is complete depends on both demand and stocking. The proxy therefore does not identify demand on censored days, where we give only a lower bound. Mechanisms A, B, and C are specified simulations rather than counterfactual inventory paths, so the transfer results measure sensitivity to these histories, not transport to another retailer. Kaplan–Meier is scored only on estimable rows, and the neural policies are budget-limited references on 600 series. The rankings depend on the locked cost ratios and the one-day shelf life, and one evaluated week cannot show seasonal effects. Any later protocol evaluated on this week would be exploratory, although the 45,000 series outside run E's sample have never been scored. The intervals use only 200 replicates. Finally, no run hashed the notebook source it executed, and the upstream dataset revision and download date are unknown, so the local file hashes are what identify the data. All runs used Linux, Python 3.11.15, two logical CPUs, and 8.42 GB of RAM; run E had no GPU.

VII. CONCLUSION

Scoring forecasts and orders against donor days makes stockout-aware comparisons on FreshRetailNet-50K testable: models trained on differently censored histories are judged on the same outcomes, under a protocol fixed before the final week was read. Calibrated demand quantiles gave the lowest cost in every train/test cell when shortage was more costly,

and a raw-sales point model with residual quantiles did so when excess stock was more costly. Moving the calibrated policy to an unseen mechanism raised its cost by at most 0.0472 normalized units. These findings hold for donor days, the simulated mechanisms, and the locked cost settings. Testing them on censored demand would need a new protocol; testing them on the 45,000 unscored series needs only a new lock.

Code and Data Availability

The notebooks, the saved run outputs, and the scripts that regenerate every table and figure without retraining are released with this paper. Run E's locked protocol is in `manifests/final_test_lock.json`. The dataset is not redistributed; it is available from Hugging Face [5] under CC BY 4.0.

The SHA-256 hashes of the local files and the digest of run E's locked protocol are:

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train.parquet  6706832db892bbae4969c19d87e07975
               d2543d2ba7d7d4756360654785de5a3d
eval.parquet   1b118840664280c6b88bffc84c80ee1f
               54c05d911e354b7599e5da10995e960e
lock digest    4ab12c4bddc06b4bb32ba8825353071b
               964ef5bfdd423bd4c876ac382180c1b6
```

References

[1] R. Fildes, S. Ma, and S. Kolassa, "Retail forecasting: Research and practice," *Int. J. Forecasting*, vol. 38, no. 4, pp. 1283–1318, 2022, doi: 10.1016/j.ijforecast.2019.06.004.

[2] S. Nahmias, "Demand estimation in lost sales inventory systems," *Naval Res. Logist.*, vol. 41, no. 6, pp. 739–757, 1994, doi: 10.1002/1520-6750(199410)41:6<739::AID-NAV3220410605>3.0.CO;2-A.

[3] W. T. Huh, R. Levi, P. Rusmevichientong, and J. B. Orlin, "Adaptive data-driven inventory control with censored demand based on Kaplan–Meier estimator," *Oper. Res.*, vol. 59, no. 4, pp. 929–941, 2011, doi: 10.1287/opre.1100.0906.

[4] Y. Wang, J. Gu, L. Long, X. Li, L. Shen, Z. Fu, X. Zhou, and X. Jiang, "FreshRetailNet-50K: A stockout-annotated censored demand dataset for latent demand recovery and forecasting in fresh retail," arXiv:2505.16319v5 [cs.LG], 2025, version 5, revised Jun. 18, 2026. [Online]. Available: https://arxiv.org/abs/2505.16319v5

[5] Dingdong-Inc, "FreshRetailNet-50K," Hugging Face dataset card, data version 1.0, 2025, accessed: Sep. 30, 2026. [Online]. Available: https://huggingface.co/datasets/Dingdong-Inc/FreshRetailNet-50K

[6] J. Azar, "A reproducible, leakage-free pipeline for censored demand forecasting and inventory optimization on FreshRetailNet-50K," *Inventions*, vol. 11, no. 5, 2026, art. no. 89, doi: 10.3390/inventions11050089.

[7] E. L. Kaplan and P. Meier, "Nonparametric estimation from incomplete observations," *J. Amer. Statist. Assoc.*, vol. 53, no. 282, pp. 457–481, 1958, doi: 10.1080/01621459.1958.10501452.

[8] O. Besbes and A. Muharremoglu, "On implications of demand censoring in the newsvendor problem," *Manage. Sci.*, vol. 59, no. 6, pp. 1407–1424, 2013, doi: 10.1287/mnsc.1120.1654.

[9] A.-L. Sachs and S. Minner, "The data-driven newsvendor with censored demand observations," *Int. J. Prod. Econ.*, vol. 149, pp. 28–36, 2014, doi: 10.1016/j.ijpe.2013.04.039.

[10] R. Kumar and O. Mouchtaki, "What is the value of censored data? An exact analysis for the data-driven newsvendor," arXiv:2602.16842 [cs.LG], 2026. [Online]. Available: https://arxiv.org/abs/2602.16842v1

[11] G.-Y. Ban and C. Rudin, "The big data newsvendor: Practical insights from machine learning," *Oper. Res.*, vol. 67, no. 1, pp. 90–108, 2019, doi: 10.1287/opre.2018.1757.

[12] D. Bertsimas and N. Kallus, "From predictive to prescriptive analytics," *Manage. Sci.*, vol. 66, no. 3, pp. 1025–1044, 2020, doi: 10.1287/mnsc.2018.3253.

[13] A. Oroojlooyjadid, L. V. Snyder, and M. Takáč, "Applying deep learning to the newsvendor problem," *IISE Trans.*, vol. 52, no. 4, pp. 444–463, 2020, doi: 10.1080/24725854.2019.1632502.

[14] R. Koenker and G. Bassett, Jr., "Regression quantiles," *Econometrica*, vol. 46, no. 1, pp. 33–50, 1978, doi: 10.2307/1913643.

[15] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017. [Online]. Available: https://papers.nips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html

[16] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, "M5 accuracy competition: Results, findings, and conclusions," *Int. J. Forecasting*, vol. 38, no. 4, pp. 1346–1364, 2022, doi: 10.1016/j.ijforecast.2021.11.013.

[17] A. Zeng, M. Chen, L. Zhang, and Q. Xu, "Are transformers effective for time series forecasting?" in *Proc. AAAI Conf. Artif. Intell.*, vol. 37, no. 9, 2023, pp. 11 121–11 128, doi: 10.1609/aaai.v37i9.26317.

[18] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, and M. Long, "TimesNet: Temporal 2D-variation modeling for general time series analysis," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023. [Online]. Available: https://iclr.cc/virtual/2023/poster/11976

[19] B. Lim, S. Ö. Arık, N. Loeff, and T. Pfister, "Temporal fusion transformers for interpretable multi-horizon time series forecasting," *Int. J. Forecasting*, vol. 37, no. 4, pp. 1748–1764, 2021, doi: 10.1016/j.ijforecast.2021.03.012.

[20] Y. Romano, E. Patterson, and E. Candès, "Conformalized quantile regression," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019. [Online]. Available: https://papers.nips.cc/paper_files/paper/2019/hash/5103c3584b063c431bd1268e9b5e76fb-Abstract.html

[21] I. Gibbs and E. Candès, "Adaptive conformal inference under distribution shift," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021. [Online]. Available: https://proceedings.neurips.cc/paper/2021/hash/0d441de75945e5acbc865406fc9a2559-Abstract.html

[22] V. Chernozhukov, I. Fernández-Val, and A. Galichon, "Quantile and probability curves without crossing," *Econometrica*, vol. 78, no. 3, pp. 1093–1125, 2010, doi: 10.3982/ECTA7880.

[23] J. Cao, "A conformal approach to feature-based newsvendor under model misspecification," arXiv:2412.13159 [cs.LG], 2024. [Online]. Available: https://arxiv.org/abs/2412.13159v1

[24] R. F. Barber, E. J. Candès, A. Ramdas, and R. J. Tibshirani, "Conformal prediction beyond exchangeability," *Ann. Statist.*, vol. 51, no. 2, 2023, doi: 10.1214/23-AOS2276.

[25] Dingdong-Inc, "frn-50k-baseline," GitHub repository, commit 8af694a, 2026, accessed: Sep. 30, 2026. [Online]. Available: https://github.com/Dingdong-Inc/frn-50k-baseline

[26] D. O. Lebedev and A. A. Romanov, "Stockout-aware segmentation and model routing for daily store-SKU demand forecasting," in *Proc. Int. Russian Smart Ind. Conf. (SmartIndustryCon)*, 2026, pp. 768–774, doi: 10.1109/SmartIndustryCon68821.2026.11492883.

[27] B. Szabłowski, "One global model, many behaviors: Stockout-aware feature engineering and dynamic scaling for multi-horizon retail demand forecasting with a cost-aware ordering policy (VN2 winner report)," arXiv:2601.18919 [cs.LG], 2026. [Online]. Available: https://arxiv.org/abs/2601.18919v1

[28] B. Efron, "Bootstrap methods: Another look at the jackknife," *Ann. Statist.*, vol. 7, no. 1, pp. 1–26, 1979, doi: 10.1214/aos/1176344552.